

Sentinel AI - Autonomous Incident Diagnosis & Resolution Agent

Hackathon Project Specification

Elevator Pitch

Sentinel AI is an AI-powered incident response platform that automatically analyzes logs, alerts, metrics, and system events to identify root causes, assess impact, recommend remediation steps, and assist engineers in resolving incidents faster.

Problem Statement

Modern systems generate application logs, infrastructure metrics, monitoring alerts, deployment events, and service health signals. During outages, engineers spend valuable time searching logs, correlating alerts, identifying root causes, determining impact, and finding remediation steps.

Solution

Sentinel AI acts as an autonomous incident investigator. Given logs, metrics, alerts, and events, it understands incidents, correlates evidence, finds probable root causes, estimates impact, suggests fixes, and generates incident reports.

Core Features

1. Incident Dashboard

Displays active incidents, severity, status, and affected services.

2. Log Upload

Supports TXT, JSON, CSV uploads or pasted logs.

3. AI Investigation Agent

Generates incident summary, root cause, evidence, confidence score, impact analysis, and remediation steps.

4. Root Cause Timeline

Visual timeline showing the chain of failure events.

5. Confidence Scoring

Ranks probable root causes with confidence percentages.

6. Impact Analysis

Estimates affected services, users, and business impact.

7. Remediation Suggestions

Provides actionable recovery recommendations.

8. Incident Report Generator

Exports investigation findings as PDF or Markdown.

AI Architecture

Agent 1: Incident Classifier

Classifies incidents as Database, Network, Application, Infrastructure, or Security.

Agent 2: Root Cause Investigator

Analyzes logs, alerts, and metrics to identify evidence-backed root causes.

Agent 3: Impact Analyzer

Determines severity, business impact, and affected systems.

Agent 4: Remediation Planner

Generates fixes, recovery plans, and runbook recommendations.

Technology Stack

Frontend: React, TypeScript, Vite, TailwindCSS, ShadCN UI

Visualization: Recharts, React Flow

Backend: FastAPI

AI Layer: Ollama + Qwen3.5:122B

Demo Scenario

Upload sample logs, trigger AI investigation, receive root cause analysis, confidence score, impact assessment, remediation recommendations, and generate an incident report.

MVP Scope

Incident Dashboard, Log Upload, AI Analysis, Root Cause Detection, Confidence Score, Remediation Suggestions, and Incident Report Generation.

Hackathon WOW Features

AI Chat with Incident, Service Dependency Graph, and Impact Simulation for proposed fixes.